

# Intelligent Vehicle Vision Systems

## 8.1 Introduction

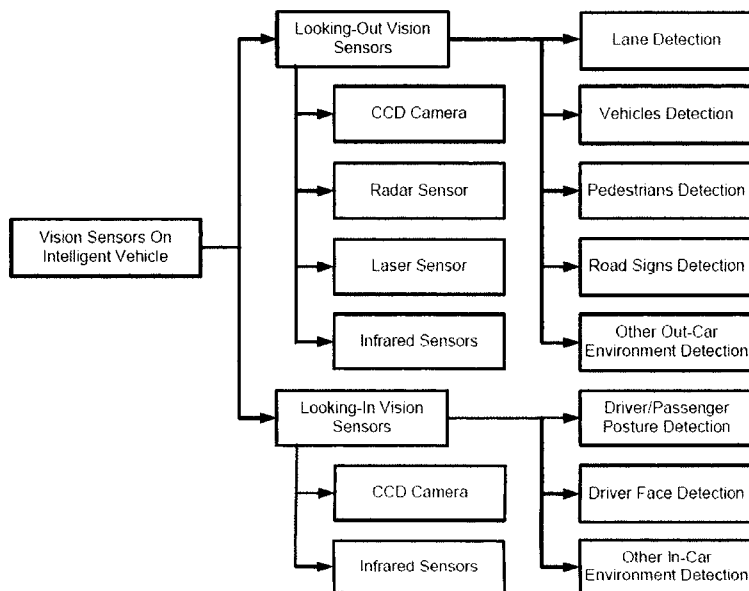
Vision sensors become increasingly important in smart transportation systems in the last twenty years. From the viewpoint of traffic infrastructure, the application of vision based traffic flow monitoring and associated control techniques offers considerable improvements over those existing methods of traffic data collection and road traffic monitoring [23]-[25]. On the other side, with the help of visual sensors, modern driver assistance system can alert the driver for dangerous situations or wrong actions, i.e. swerving off lane/road or disregarding traffic signs/lights; or even autonomously take the control of vehicle. Now, vision based driving safety system is world- widely viewed as an essential component of the next generation vehicles [1]-[22].

During the last three decades, vision based driving safety research has attracted great interests and resulted in a wealth of promising results. From their functions, varied vision sensors can be divided into two types: “looking-in” sensors that detect out-vehicle environment and “looking-out” sensors that monitor in-vehicle environment; Fig.8.1.

CMOS/CCD cameras are the most frequently used vision sensors in intelligent vehicle. As pointed out in many literals, CMOS/CCD cameras are superior to many other sensors due to their fast response, easy installation/operation/maintenance, and ability to monitor wide areas. But CMOS/CCD cameras perform perfect only when they work under good environment conditions. In real scenarios, shadows of trees/clouds and dim lighting etc. prevent us from obtaining clear contrasting digital images occasionally.

Recently, applications of mm-wave radar, laser and infrared sensors as alternatives to cameras got significant considerations. One important merit of these new sensors is that they are not sensitive to weather/lighting conditions. Although CMOS/CCD cameras cannot be thoroughly alternated, they are outperformed by these new sensors in lots of applications. In this

Chapter, these new sensors are also viewed as general vision sensors and discussed. And the sensor fusion between different vision sensors is addressed, too.



**Fig.8.1** Divisions and functions of “looking-in” and “looking out” intelligent vision sensors.

As mentioned in [9], [22], the basic visual sensing tasks of an intelligent vehicle can be outlined as follows:

out-vehicle vision detection:

- (1) extracting lane boundaries, especially when they are not clearly marked and/or in bad weather conditions;
- (2) detecting vehicles nearby, estimating their positions, speeds, and accelerations;
- (3) recognizing the relevant traffic signs and traffic lights;
- (4) detecting the unexpected traffic participants/obstacles, such as pedestrians and big stones;

in-vehicle detection:

- (5) monitor the driver’s eyes movement for fatigue detection;